

where $C_q^\pm(\widetilde{X}) = \{c \in C_q(\widetilde{X}) \mid t_\#(c) = \pm c\}$. Obviously, $C_q^+(\widetilde{X})$ is the same as $C_q(X)$; we take the other summand, $C_q^-(\widetilde{X})$, for $C_q(X; \mathbb{Z}_T)$. The groups $C_q(X; \mathbb{Z}_T)$ form, in the obvious way, a complex. The homology of this complex is denoted as $H_q(X; \mathbb{Z}_T)$, and the corresponding cohomology is taken for $H^q(X, \mathbb{Z}_T)$. We will not discuss in any detail these homology and cohomology with “twisted coefficients”; moreover, we will have to do it in a much bigger generality in Chap. 3. Now we restrict ourselves to a recommendation to the reader to reconstruct Poincaré isomorphism given above [they are cap-products with a “fundamental class” $[X] \in H_n(X; \mathbb{Z}_T)$], and Poincaré duality with appropriately defined intersection numbers and secondary intersection numbers.

Lecture 18 The Obstruction Theory

18.1 Obstructions to Extending a Continuous Map

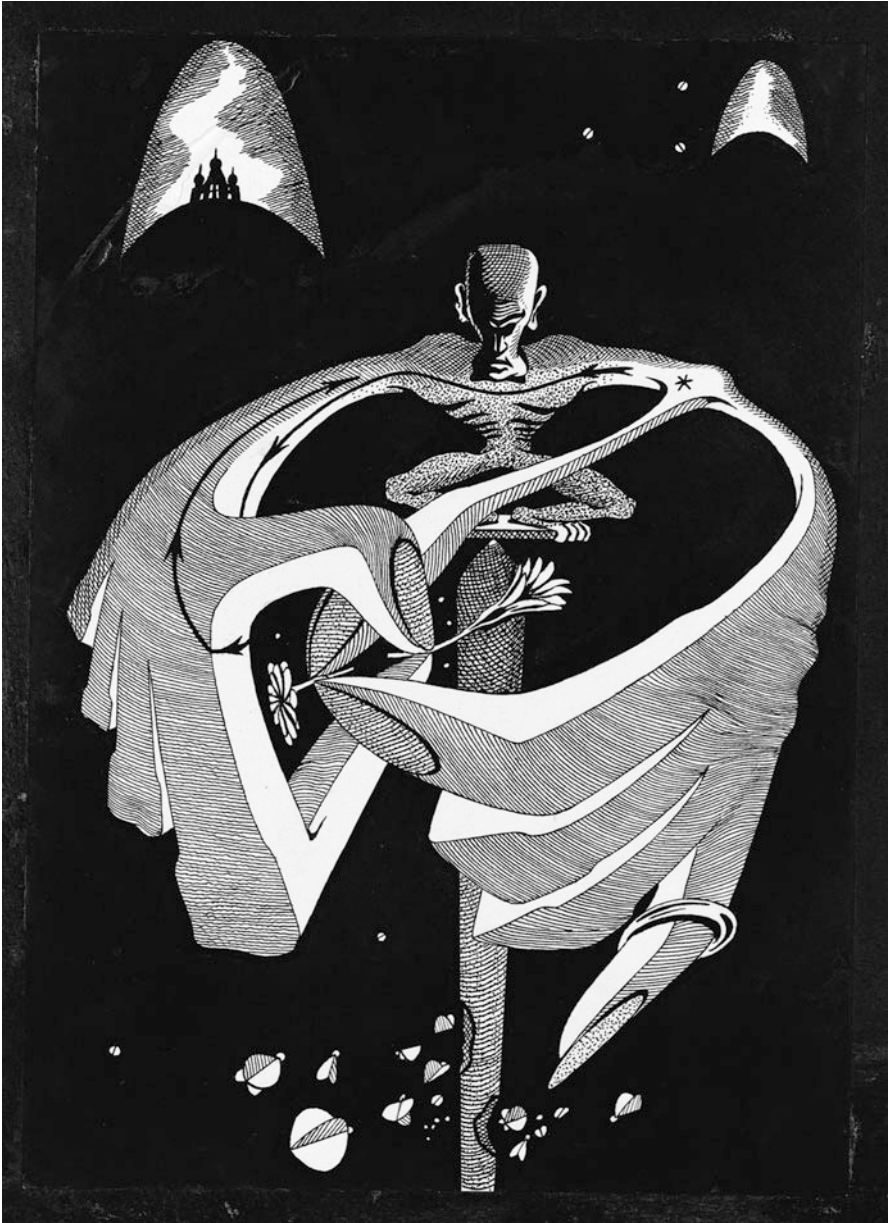
Most problems in homotopy topology consist in a homotopy classification of continuous maps between two topological spaces. A natural intermediate problem is the question of whether a given continuous map $A \rightarrow Y$ can be extended to a continuous map $X \rightarrow Y$ for some $X \supset A$ (with a subsequent classification of such extensions). This is what the obstruction theory was designed for. We will begin with a technically important particular case.

Let X be a CW complex, and let Y be a connected topological space which is assumed homotopically simple (that is, the action of the fundamental group in all homotopy groups is trivial; later, we will discuss several possibilities of removing or, at least, weakening this condition). Consider the problem of extending a continuous map $f: X^n \rightarrow Y$ to a continuous map $X^{n+1} \rightarrow Y$ (where X^n, X^{n+1} are skeletons). Let $e \subset X$ be a cell of dimension $n + 1$, and let $h: D^{n+1} \rightarrow X$ be a corresponding characteristic map. There arises a continuous map $f_e = f \circ h|_{S^n}: S^n \rightarrow Y$. It is obvious that f can be continuously extended to $X^n \cup e$ if and only if f_e is homotopic to a constant, that is, if f_e represents the class $0 \in \pi_n(Y)$ (since Y is homotopically simple, we do not need to fix a base point in Y).

Furthermore, the possibility of extension of f to X^{n+1} is the same as the possibility of its extension to every $(n + 1)$ -dimensional cell of X . If we construct, as above, a map $f_e: S^n \rightarrow Y$ for every e and denote by φ_e the class of f_e in $\pi_n(Y)$, we arrive at the following, essentially tautological, statement: A continuous map $f: X^n \rightarrow Y$ can be extended to a continuous map $X^{n+1} \rightarrow Y$ if and only if every φ_e is equal to 0.

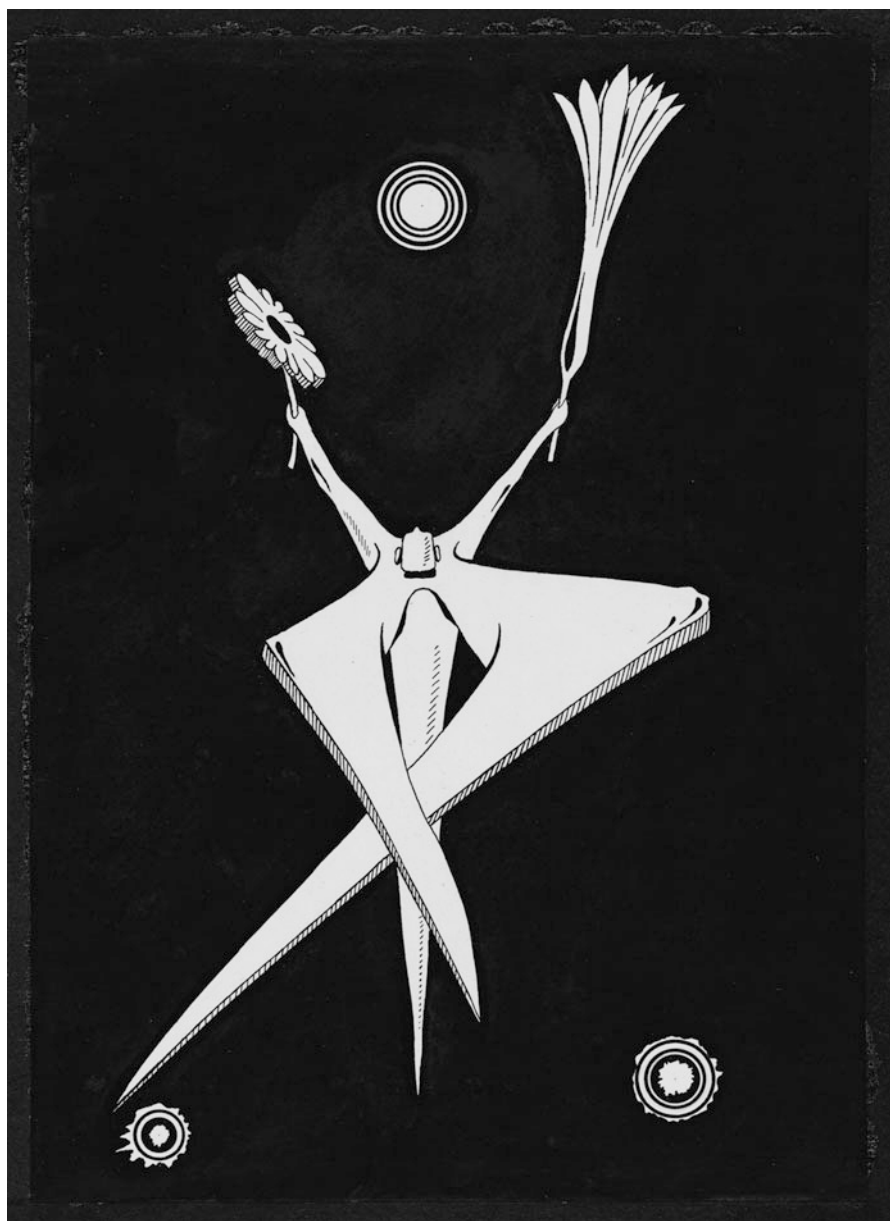
The function $e \mapsto \varphi_e$ can be regarded as an $(n + 1)$ -dimensional cellular cochain c_f of X with coefficients in $\pi_n(Y)$. (This cochain does not depend on the choice of characteristic maps. Indeed, from the homotopy point of view there are only two characteristic maps corresponding to the two orientations of e ; the replacement of h











by a characteristic map of the opposite orientation changes the sign at φ_e , but also reverses the orientation of e , so the cochain c_f stays unchanged.) Thus,

$$c_f \in \mathcal{C}^{n+1}(X; \pi_n(Y)),$$

and f can be extended to X^{n+1} if and only if $c_f = 0$. The cochain c_f is called the *obstruction cochain* to the extension of f to X^{n+1} .

Notice that the obstruction cochains have a naturality property: If $\varphi: X' \rightarrow X$ is a cellular map and $\psi: Y \rightarrow Y'$ is a continuous map, then $c_{\psi \circ \varphi} = \varphi^\# \psi_\# c_f$.

Up to now, everything said was a sheer triviality. Here is the first nontrivial statement.

Theorem 1. *The obstruction cochain is a cocycle: $\delta c_f = 0$.*

Proof. The statement may be regarded as a variation on the theme of $\partial\partial = 0$ [we need to prove that $c_f(\partial a) = 0$, but the cochain c_f itself is defined by means of boundaries], but the accurate proof requires some work. For example, it can be deduced from the relative Hurewicz theorem (Sect. 14.4). According to this theorem, if X satisfies some conditions (we will discuss them later), then the Hurewicz homomorphism $h: \pi_q(X^q, X^{q-1}) \rightarrow H_q(X^q, X^{q-1})$ is an isomorphism. Consider the diagram

$$\begin{array}{ccc}
 \mathcal{C}_{n+2}(X) = H_{n+2}(X^{n+2}, X^{n+1}) & \xrightarrow{h^{-1}} & \pi_{n+2}(X^{n+2}, X^{n+1}) \\
 \downarrow \partial & & \downarrow \partial \\
 & & \pi_{n+1}(X^{n+1}) \\
 & & \downarrow \\
 \mathcal{C}_{n+1}(X) = H_{n+1}(X^{n+1}, X^n) & \xrightarrow{h^{-1}} & \pi_{n+1}(X^{n+1}, X^n) \\
 \searrow c_f & & \downarrow \partial \\
 & & \pi_n(X^n) \\
 & & \downarrow f_* \\
 & & \pi_n(Y)
 \end{array}
 \left. \vphantom{\begin{array}{c} \pi_{n+1}(X^{n+1}) \\ \pi_{n+1}(X^{n+1}, X^n) \\ \pi_n(X^n) \end{array}} \right\} 0$$

This diagram is commutative by the definition of the cochain c_f and the homomorphism $\partial: \mathcal{C}_{n+2}(X) \rightarrow \mathcal{C}_{n+1}(X)$. Also, the part of the vertical column marked by a brace is a fragment of the homotopy sequence of the pair (X^{n+1}, X^n) , and hence the composition of homomorphism within this part is 0. Thus, $c_f \circ \partial = \delta c_f = 0$.

However, the reference to the relative Hurewicz theorem forces us to respect its assumptions, that is, to assume that X is simply connected and that $n + 1 > 1$.

We will ignore the second assumption (it is easy to see that our arguments are valid when $n = 0$), and we can get rid of the simply connectedness assumption in the following way. Let $p: \tilde{X} \rightarrow X$ be the universal covering of X . The CW decomposition of X induces a CW decomposition of $\tilde{X}$, and the map $p^\#: C^q(X) \rightarrow C^q(\tilde{X})$ is a monomorphism. For a map $f: X^n \rightarrow Y$, the obstruction cochain $c_{f \circ p} \in C^{n+1}(\tilde{X}; \pi_n(Y))$ is $p^\# c_f$, $p^\# \delta c_f = \delta p^\# c_f = \delta c_{f \circ p} = 0$, and hence $\delta c_f = 0$.

The cohomology class $C_f \in H^{n+1}(X; \pi_n(Y))$ of the cocycle c_f is called the *cohomology obstruction*, or simply the *obstruction to extension of f to X^{n+1}* .

Theorem 2. *The condition $C_f = 0$ is necessary and sufficient to the existence of extending $f|_{X^{n-1}}$ to X^{n+1} . In other words, $C_f = 0$ if and only if it is possible to extend f to X^{n+1} after, possibly, a changing f on $X^n - X^{n-1}$.*

[One can apply this theorem to successive extensions of f from a skeleton to a skeleton. Say, let us have a continuous map $f: X^n \rightarrow Y$. There arises an obstruction $C_f \in H^{n+1}(X; \pi_n(Y))$. If it is 0, we can extend f to X^{n+1} at the price of some modification of f on X^n not touching f on X^{n-1} . In this case (that is, if $C_f = 0$), we get a new obstruction in $H^{n+2}(X; \pi_{n+1})$. If it is zero, we extend f to X^{n+2} (maybe, after changing the previous extension), and get the next obstruction in $H^{n+3}(X; \pi_{n+2}(Y))$, and so on. One should remember, however, that every new obstruction depends from the previous extension, and hence these obstructions are defined with a growing indeterminacy.]

Before proving Theorem 2, we will give a new definition which will be useful in the proof but will also have a considerable independent value. Let $f, g: X^n \rightarrow Y$ be two continuous maps which agree on X^{n-1} . Consider an arbitrary n -dimensional cell e with a characteristic map $h: D^n \rightarrow X$. The maps $f \circ h, g \circ h: D^n \rightarrow Y$ agree on S^{n-1} [since $h(S^{n-1}) \subset X^{n-1}$, and f and g agree on X^{n-1}] and together compose a map $k_e: S^n \rightarrow Y$ (which is $f \circ h$ on the lower hemisphere and $g \circ h$ on the upper hemisphere). We define the *difference cochain*

$$d_{f,g} \in C^n(X; \pi_n(Y)),$$

whose value on e is the class of k_e in $\pi_n(Y)$. It is clear that the condition $d_{f,g} = 0$ is necessary and sufficient for the existence of a homotopy between f and g which is fixed on X^{n-1} (in the terminology of Chap. 1, an X^{n-1} -homotopy; see Sect. 5.7). In the important case when f and g are defined on the whole X and agree on X^{n-1} , the condition $d_{f,g} = 0$ is necessary and sufficient for the existence of an X^{n-1} -homotopy of f making f agree with g on X^n (for this statement, we need to use Borsuk's theorem, Sect. 5.5). Notice also that the difference cochains have a naturality property similar to that of the obstruction cochains: $d_{\psi \circ f \circ \varphi, \psi \circ g \circ \varphi} = \varphi^\# \psi_\# d_{f,g}$.

Lemma 1. *For any continuous map $f: X^n \rightarrow Y$ and any cochain $d \in C^n(X; \pi_n(Y))$, there exists a continuous map $g: X^n \rightarrow Y$ which agrees with f on X^{n-1} and is such that $d_{f,g} = d$.*

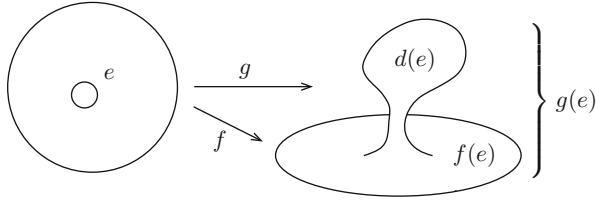


Fig. 74 Proof of Lemma 1

Proof. Consider an n -dimensional cell e of X and distinguish a small ball in e . Then change the map f on this ball in such a way that the two maps of the ball, the old one and the new one, compose a spheroid of the class $d(e)$ (see Fig. 74). Having such a change made on each n -dimensional cell, we get the map g with the required properties.

Lemma 2. $\delta d_{f,g} = c_g - c_f$.

Proof. Consider, for simplicity's sake, the case when f and g are different on only one n -dimensional cell $e \subset X$ (the general case, essentially, is not different from this case). Let σ be an $(n+1)$ -dimensional cell of X ; we want to show that

$$c_g(\sigma) - c_f(\sigma) = [\sigma : e] d_{f,g}(e).$$

Let $h: D^{n+1} \rightarrow X$ be a characteristic map for σ . We can assume that $h^{-1}(e)$ consists of several open balls, of which every one is mapped by h homeomorphically onto e , with preserving or reversing the orientation, and $[\sigma : e]$ is the difference of the number of balls where the orientation is preserved and the number of balls where it is reversed (compare the description of the incidence numbers in Sect. 13.6). This makes the desired equality obvious: A spheroid representing $c_g(\sigma)$ is obtained from a spheroid representing $c_f(\sigma)$ by adding spheroids of the class $\pm d_{f,g}(e)$, and the algebraic number of these spheroids is $[\sigma : e]$.

Proof of Theorem 2. If $C_f = 0$, then $c_f = \delta d$ and, by Lemma 1, there exists a map $g: X^n \rightarrow Y$ such that $g|_{X^{n-1}} = f|_{X^{n-1}}$ and $d_{f,g} = -d$. But then, by Lemma 2, $c_g = c_f + \delta d_{f,g} = \delta d - \delta d = 0$; thus, g can be extended to X^{n+1} . Conversely, if there exists a map $g: X^n \rightarrow Y$ which agrees with f on X^{n-1} and can be extended to X^{n+1} , then $c_g = 0$ and $c_f = c_g - \delta d_{f,g} = \delta d_{f,g}$, and hence $C_f = 0$.

Remark. The two lemmas of this proof are not less important than the theorem; we will use them later.

EXERCISE 1. Prove that $d_{g,f} = -d_{f,g}$ and $d_{f,h} = d_{f,g} + d_{g,h}$.

18.2 The Relative Case

Let A be a CW subcomplex of a CW complex X , and let the continuous map f be defined on $A \cup X^n$. The obstruction cochain c_f to an extension of this map to $A \cup X^{n+1}$ is contained in $\mathcal{C}^{n+1}(X, A; \pi_n(Y))$, it is a cocycle, and its cohomology class $C_f \in H^{n+1}(X, A; \pi_n(Y))$ is called an obstruction. The theory of these relative obstructions is absolutely parallel to its absolute prototype; in particular, it contains the notion of difference cochains, and there are precise analogies (for both the statements and the proofs) of all theorems and lemmas of the previous section. We will point out the following important consequence of the relative theory in the absolute theory.

Let $f, g: X \rightarrow Y$ (or $X^{n+1} \rightarrow Y$) be two maps with $f|_{X^{n-1}} = g|_{X^{n-1}}$ [or with a fixed homotopy connecting $f|_{X^{n-1}}$ and $g|_{X^{n-1}}$]. We consider the problem of constructing a homotopy between f and g fixed (or coinciding with the given homotopy) on X^{n-1} . This problem is equivalent to extending to $X \times I$ (or to $X^{n+1} \times I$) the map which is given on $(X \times 0) \cup (X^{n-1} \times I) \cup (X \times 1)$ by the formula

$$(x, t) \mapsto \begin{cases} f(x), & \text{if } t = 0 \text{ or } x \in X^{n-1}, \\ g(x), & \text{if } t = 1 \text{ (or } x \in X^{n-1}) \end{cases}$$

(this formula is for the case when f and g agree on X^{n-1} ; if a homotopy between $f|_{X^{n-1}}$ and $g|_{X^{n-1}}$ is given, the formula will be slightly different; we leave the details to the reader). The obstruction to an extension of this map to $(X \times 0) \cup (X^n \times I) \cup (X \times 1)$ lies in $\mathcal{C}^{n+1}(X \times I, (X \times 0) \cup (X \times 1); \pi_n(Y)) = \mathcal{C}^n(X; \pi_n(Y))$, and it is easy to see that it is nothing but $d_{f,g}$. By the way, $\delta d_{f,g} = c_g - c_f = 0$, since f and g are both defined on the whole X (or, at least, on X^{n+1}). If we apply to this situation the relative version of Theorem 2 of Sect. 18.1, we will get the following result.

Theorem. *If $f, g: X \rightarrow Y$ are two continuous maps which agree on X^{n-1} , then the difference cochain $d_{f,g}$ is a cocycle whose cohomology class $D_{f,g} \in H^n(X; \pi_n(Y))$ is equal to 0 if and only if $f|_{X^n}$ and $g|_{X^n}$ are X^{n-2} -homotopic.*

18.3 The First Application: Cohomology and Maps into $K(\pi, n)$ s

The main result of this section was promised in Lecture 4. Let π be an Abelian group.

Recall that the construction of a $K(\pi, n)$ space begins with taking a bouquet of n -dimensional spheres set into a correspondence with some system of generators of π (see Sect. 11.7); then we attach to this bouquet cells of dimensions $> n$. If we assign to every n -dimensional cell of $K(\pi, n)$ the corresponding element of π , we get a cochain $c \in \mathcal{C}^n(K(\pi, n); \pi)$ [we admit here a certain abuse of notation, using the symbol $K(\pi, n)$ for a CW complex obtained by some concrete construction].

Lemma. c is a cocycle.

First Proof (Direct). The cells of dimension $n + 1$ correspond to the defining relations between the chosen generators. If the cell σ corresponds to the relation $\sum k_i g_i = 0$ between the generators g_i , then for the n -dimensional cell e_i corresponding to the generator g_i , the incidence number $[\sigma : e_i]$ is k_i . Then

$$\delta c(\sigma) = \sum_i [\sigma : e_i] c(e_i) = \sum_i k_i g_i = 0.$$

Second Proof (Indirect). Actually, $c = d_{\text{const}, \text{id}}$; thus, $\delta c = 0$ by Lemma 2 of Sect. 18.1.

The cohomology class $F_\pi \in H^n(K(\pi, n); \pi)$ of the cocycle c is called the *fundamental cohomology class* of $K(\pi, n)$. Another description of this class: According to the universal coefficients formula,

$$H^n(K(\pi, n); \pi) = \text{Hom}(H_n(K(\pi, n)), \pi),$$

and, by Hurewicz's theorem, $H_n(K(\pi, n)) = \pi_n(K(\pi, n)) = \pi$. The class F_π corresponds to the identity homomorphism

$$\text{id}_\pi \in \text{Hom}(H_n(K(\pi, n)), \pi).$$

EXERCISE 2. Prove the equivalence of the two definitions of the fundamental homology class.

Notice that the second definition of the fundamental class can be applied to an arbitrary $(n - 1)$ -connected space X . In this case, it yields a cohomology class $F_X \in H^n(X; \pi_n(X))$. We will return to this class later.

Now we turn to the main result of this section.

Theorem. *Let X be a CW complex. For any Abelian group π and for any $n > 0$, the map*

$$\pi(X, K(\pi, n)) \rightarrow H^n(X; \pi), \quad [f] \rightarrow f^*(F_\pi), \quad (*)$$

is a bijection.

Proof. First, let $\gamma \in H^n(X; \pi)$, and let $c \in C^n(X; \pi)$ be a cocycle of the class γ . We want to construct a continuous map $f: X \rightarrow K(\pi, n)$ which takes the cocycle of class F_π (constructed above) into c . By Lemma 1 of Sect. 18.1, there exists a map $f: X^n \rightarrow K(\pi, n)$ such that $f(X^{n-1})$ is the (only) vertex of $K(\pi, n)$ and $d_{\text{const}, f} = c$. Then, obviously, $f^\#: C^n(K(\pi, n); \pi) \rightarrow C^n(X; \pi)$ takes $d_{\text{const}, \text{id}}$ into $d_{\text{const}, f} = c$ (by the naturality property of the difference cochains; see Sect. 18.1). Then we extend this map f to $X^{n+1}, X^{n+2}, \dots$, and it is possible, since $\pi_{n+1}(K(\pi, n)), \pi_{n+2}(K(\pi, n)), \dots$ are all zeroes. We obtain a map $f: X \rightarrow K(\pi, n)$. By construction, f^* takes F_π into γ . Thus, the map $(*)$ is onto.

Now let $f, g: X \rightarrow K(\pi, n)$ be two continuous maps with $f^*F_\pi = g^*F_\pi$. We want to prove that $f \sim g$; we can assume that f and g are cellular maps (in particular, they are constant on X^{n-1}). Then f^*F_π and g^*F_π are represented by $f^\#d_{\text{const}, \text{id}} = d_{\text{const}, f}$ and $g^\#d_{\text{const}, \text{id}} = d_{\text{const}, g}$. Hence, the cocycles $d_{\text{const}, f}$ and $d_{\text{const}, g}$ are cohomological, so the difference $d_{\text{const}, g} - d_{\text{const}, f} = d_{f, g}$ is cohomological to 0, or $D_{f, g} = 0$. According to the theorem in Sect. 18.2, this shows that f and g are X^{n-2} -homotopic (the homotopy being fixed on X^{n-2} is not important to us) on X^n . They are also homotopic on further skeletons, since the further difference cochains belong to the cochain groups with trivial coefficients. Thus, the map $(*)$ is one-to-one.

Corollary 1. *A CW complex of the type $K(\pi, n)$ is homotopically unique. Hence, a topological space of the type $K(\pi, n)$ is weakly homotopically unique.*

Proof. Let X, X' be CW complexes of the type $K(\pi, n)$, and let $F_\pi \in H^n(X; \pi)$, $F'_\pi \in H^n(X'; \pi)$ be the fundamental classes. According to the theorem, there exist continuous maps $f: X \rightarrow X'$, $g: X' \rightarrow X$ such that $f^*(F'_\pi) = F_\pi$ and $g^*(F_\pi) = F'_\pi$. Since $(g \circ f)^*(F_\pi) = f^* \circ g^*(F_\pi) = F_\pi = (\text{id}_X)^*(F_\pi)$, we have $g \circ f \sim \text{id}_X$ and, similarly, $f \circ g \sim \text{id}_{X'}$.

EXERCISE 3. Since $K(\pi, n) \sim \Omega K(\pi, n+1)$ is an H -space, the set $\pi(X, K(\pi, n))$ is a group (see Lecture 4), and the bijection $H^n(X; \pi) \leftrightarrow \pi(X, K(\pi, n))$ is a bijection between two groups. Prove that it is a group isomorphism.

Actually, for every Abelian group π and every n , there exists an Abelian topological group of the type $K(\pi, n)$. The reader may try to prove it by an appropriate enhancing of the construction of the (second) loop space.

Corollary 2. *For a CW complex X , there is a group isomorphism $H^1(X; \mathbb{Z}) \cong \pi(X, S^1)$ (where S^1 is regarded as an Abelian topological group).*

EXERCISE 4. Prove that every continuous map $\underbrace{S^1 \times \cdots \times S^1}_n \rightarrow \underbrace{S^1 \times \cdots \times S^1}_m$ is homotopic to a linear map (that is, to a map obtained by a factorization from a linear map $\mathbb{R}^n \rightarrow \mathbb{R}^m$ determined by an integral matrix).

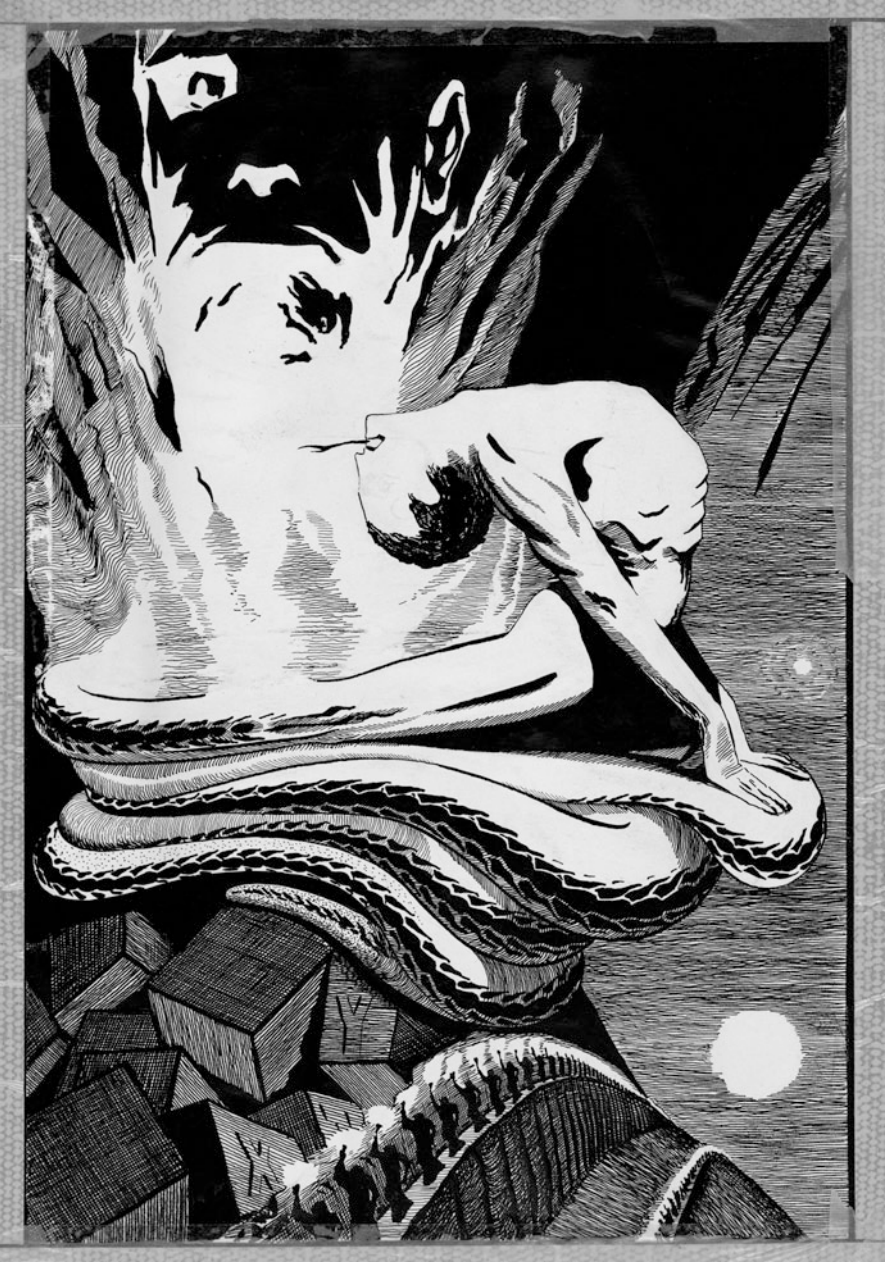
18.4 The Second Application: Hopf's Theorems

Theorem 1 (Hopf). *For every n -dimensional CW complex X , there is a bijection*

$$H^n(X; \mathbb{Z}) \leftrightarrow \pi(X, S^n), [f] \mapsto f^*(s),$$

where $s = 1 \in \mathbb{Z} = H^n(S^n; \mathbb{Z})$.

Proof. This classical theorem (proved, actually, before the appearance of not only the obstruction theory, but also cohomology) is, from a modern point of view, a corollary of the theorem in Sect. 18.3. Indeed, the construction of the space



$K(\pi, n)$, as given in Sect. 11.7, begins with a bouquet of n -dimensional spheres corresponding to generators of π ; if $\pi = \mathbb{Z}$, we can take one sphere. On the next step, we attach $(n + 1)$ -dimensional cells corresponding to relations between the chosen generators; but in the case $\pi = \mathbb{Z}$ there are no relations, and no $(n + 1)$ -dimensional cells are needed. Then we attach cells of dimensions $\geq n + 2$. We see that the $(n + 1)$ st skeleton of (such constructed) $K(\mathbb{Z}, n)$ is S^n . Hence, by the cellular approximation theorem, if X is n -dimensional, every map $X \rightarrow K(\mathbb{Z}, n)$ is homotopic to a map $X \rightarrow S^n \subset K(\mathbb{Z}, n)$ and every two maps $X \rightarrow S^n \subset K(\mathbb{Z}, n)$ homotopic in $K(\mathbb{Z}, n)$ are homotopic in S^n .

[There is a more direct proof which is a replica of the proof of the theorem in Sect. 18.3. The main difference is that the higher obstruction and difference cochains are equal to zero not because the higher homotopy groups of S^n are zeroes (which is not true), but because X has no cells of higher dimensions.]

Theorem 2 (Hopf). *Let an n -dimensional CW complex X contain as a CW sub-complex a sphere S^{n-1} . This sphere is a retract of X if and only if the inclusion homomorphism $H^{n-1}(X; \mathbb{Z}) \rightarrow H^{n-1}(S^{n-1}; \mathbb{Z})$ is an epimorphism.*

Proof. The only if part is obvious: If $r: X \rightarrow S^{n-1}$ is a retraction, then the composition

$$H^{n-1}(S^{n-1}; \mathbb{Z}) \xrightarrow{r^*} H^{n-1}(X; \mathbb{Z}) \xrightarrow{j^*} H^{n-1}(S^{n-1}; \mathbb{Z}),$$

where j is the inclusion map, is the identity, and hence j^* is an epimorphism. Assume now that j^* is an epimorphism and fix a class $\alpha \in H^{n-1}(X; \mathbb{Z})$ such that $j^*(\alpha) = 1 \in \mathbb{Z} = H^{n-1}(S^{n-1}; \mathbb{Z})$. Let $a \in C^{n-1}(X; \mathbb{Z})$ be a cocycle of the class α . Construct a map $q: X \rightarrow S^{n-1}$ in the following way. All the cells of dimensions $\leq n - 2$ we map into a point. On every $(n - 1)$ -dimensional cell e define the map as the spheroid of the class $a(e)$. This requirement means precisely that the map $q^\#$ takes $1 \in \mathbb{Z} = C^{n-1}(S^{n-1}; \mathbb{Z})$ into a . On the other side, it means that the cochain a is the difference cochain between the already constructed part of the map q and the map $\text{const}: X^{n-1} \rightarrow S^{n-1}$. Hence,

$$0 = \delta a = \delta d_{q, \text{const}} = c_q - c_{\text{const}} = c_q,$$

so the map q can be extended to $X^n = X$. The composition

$$S^{n-1} \xrightarrow{j} X \xrightarrow{q} S^{n-1}$$

induces the identity map in cohomology: $(q \circ j)^*(1) = j^*(q^*(1)) = j^*(\alpha) = 1$, and hence homotopic to id. We can extend the homotopy between this map and id to the homotopy of the map q . As a result, we will get a map $r: X \rightarrow S^{n-1}$ which is the identity on S^{n-1} , that is, a retraction.

18.5 Obstructions to Extensions of Sections

Let $\xi = (E, B, F, p)$ be a locally trivial fibration. We assume that the fiber F is homotopically simple (for example, simply connected), and the base B is simply connected. (The last assumption can be weakened to the assumption of the *homotopical simplicity of the fibration*. The latter means that for every continuous map $S^1 \rightarrow B$, the induced fibration over S^1 is trivial. In the next lecture, we will encounter important examples of this situation.)

Assume that the base B is a CW complex and that there given a section $s: B^n \rightarrow E$ [which means that $p \circ s = \text{id}$] over the n th skeleton of the base. We are going to describe an obstruction to extending this section to B^{n+1} . Let e be an $(n+1)$ -dimensional cell over B . The fibration $h^*\xi$ over D^{n+1} , induced by means of a characteristic map $h: D^{n+1} \rightarrow B$ for the cell e , is trivial. The section s induces a section $S^n \rightarrow D^{n+1} \times F$ of the restriction of the last fibration to $S^n \subset D^{n+1}$, and hence an element of $\pi_n(D^{n+1} \times F) = \pi_n(F)$ (rather of the fiber $p^{-1}(x)$ over some point $x \in e$, but the simply connectedness of the base, or the homotopical simplicity of the fibration ξ , provides a canonical homomorphism between homotopy groups of all fibers—the reader will reconstruct a detailed explanation of this). We get a cochain $c_s \in \mathcal{C}^{n+1}(B; \pi_n(F))$. This is the *obstruction cochain to extending s to B^{n+1}* . The properties of this obstruction cochain are the same as those of the obstruction cochains considered in Sect. 18.1. Namely:

- (1) The section s can be extended to a section over the $(n+1)$ st skeleton of B if and only if $c_s = 0$.
- (2) $\delta c_s = 0$.
- (3) The cohomology class $C_s \in H^{n+1}(B; \pi_n(F))$ of c_s (which is called the *obstruction*) is equal to 0 if and only if the section s can be extended to a section over B^{n+1} .

There are also *difference cochains* $d_{s,s'}$ whose definition and properties are the same as before.

Obstructions to extending maps may be regarded as particular cases of obstructions to extending sections. Namely, a continuous map $f: X \rightarrow Y$ can be represented by the graph $F: X \rightarrow X \times Y$, $F(x) = (x, f(x))$, which, in turn, is a section of the trivial fibration $(X \times Y, X, Y, p)$, where $p: X \times Y \rightarrow X$ is the projection of the product onto a factor. Obstructions to extending a map are the same as obstructions to extending its graph. On the other hand, the theory of obstructions to sections cannot be reduced to the theory of obstructions to maps. In particular, the latter does not have any analogy of the next construction.

Suppose that $\pi_0(F) = \pi_1(F) = \dots = \pi_{n-1}(F) = 0$, and $\pi_n(F) \neq 0$. Then there are no obstructions to extending a section from B^0 (where it obviously exists) to $B^1, \dots, B^{n-1}$ and the first obstruction emerges in $H^{n+1}(B; \pi_n(F))$: It is the obstruction to extending the section from B^{n-1} to B^n . This obstruction could depend, however, on the sections on the previous skeletons; however, the next proposition states that it is not the case.

Proposition 1. *Let $\pi_0(F) = \pi_1(F) = \cdots = \pi_{n-1}(F) = 0$, and let $s, s': B^n \rightarrow E$ be two sections. Then $C_s = C_{s'} \in H^{n+1}(B; \pi_n(F))$.*

To prove this, we need a slightly modified version of the homotopy extension property (Borsuk's theorem; see Sect. 5.5).

Lemma (Borsuk's Theorem for Sections). *Let $\xi = (E, B, F, p)$ be a locally trivial fibration with a CW base, let $S: B \rightarrow E$ be a section of ξ , let A be a CW subspace of B , and let $s_i: A \rightarrow E$ be a homotopy consisting of sections of $\xi|_A$ such that $s_0 = S|_A$. Then there exists a homotopy $S_t: B \rightarrow E$ consisting of sections of ξ and such that $S_0 = S$, $S_t|_A = s_t$.*

Proof of Lemma. This lemma is not different from Borsuk's theorem in the case when the fibration is (standard) trivial: $E = B \times F$, p is the projection of the product onto a factor. Indeed, in this case, a section is the same as a continuous map $B \rightarrow F$. Passing to the general case, we can restrict ourselves to the situation when A and B differ by one cell: $B = A \cup e$, where e is a cell of B . Take a characteristic map $h: D^n \rightarrow B$ (where $n = \dim e$). Then the sections S, s_t of ξ and $\xi|_A$ give rise to sections S', s'_t of the fibrations $h^*\xi, h^*\xi|_{S^{n-1}}$ [such that $s'_0 = S'|_{S^{n-1}}$]. Since the fibration $h^*\xi$ is trivial (Feldbau's theorem, Sect. 9.2), the lemma has already been proved for this fibration, which provides a homotopy S'_t consisting of sections of this fibration such that $S'_0 = S'$ and $S'_t|_{S^{n-1}} = s'_t$. The homotopies s_t and S'_t together form a homotopy $S_t: B \rightarrow E$ with the required properties.

Proof of Proposition 1. It is clear that a homotopy of a section $s: B^k \rightarrow E$ will not affect either c_s or C_s . Suppose that the given sections s, s' are homotopic over B^k for some k , $0 \leq k < n-1$ (since the fiber F is connected, this is obviously true for $k = 0$). A homotopy of s' to s on B^k can be extended, by the lemma, to a homotopy of s' on B^n , without any changes for $c_{s'}$ and $C_{s'}$ so we can assume that $s' = s$ on B^k . The difference cochain $d_{s,s'} \in C^{k+1}(B; \pi_{k+1}(F))$ is zero, because $\pi_{k+1}(F) = 0$; thus, $s' \sim s$ on B^{k+1} . In this way, we can reduce the general case of the proposition to the case when $s' = s$ on B^{n-1} . Then we have a difference cochain $d_{s,s'} \in C^n(B, \pi_n(X))$, and $\delta d_{s,s'} = c_{s'} - c_s$. Thus, the cocycles c_s and $c_{s'}$ are cohomological and hence $C_s = C_{s'}$.

Proposition 1 shows that the *first obstruction* to extending a section to the n th skeleton of the base is determined by the fibration, so we obtain a well-defined class $C(\xi) \in H^{n+1}(B; \pi_n(F))$ (recall that n is the number of the first nontrivial homotopy group of F); this class is called the *characteristic class* of ξ ; we will also use the term *primary characteristic class* to distinguish it from numerous characteristic classes of vector bundles, which will be studied in Lecture 19.

One can say that a fibration as above has a section over the n th skeleton of the base if and only if its characteristic class is zero.

EXERCISE 5 (The main property of characteristic classes). Let ξ be a fibration as above, and let $f: B' \rightarrow B$ be a continuous map of some CW complex into B . Then

$$C(f^*\xi) = f^*(C(\xi)).$$

EXERCISE 6. Prove that a characteristic class is homotopy invariant (we leave to the reader not only the proof, but also a precise statement of this fact). In particular, the characteristic class does not depend on the CW structure of the base.

EXERCISE 7. Using previous exercises, make up a definition of a characteristic class in the case when the base is not a CW complex.

Example. (Since this example concerns smooth manifolds, the definitions and statements will not be genuinely rigorous.) Let X be a connected closed oriented n -dimensional manifold and let T be the manifold of all nonzero tangent vectors of M . The projection $p: T \rightarrow X$ (which assigns to a tangent vector the tangency point) gives rise to a locally trivial fibration $\tau_X = (T, X, \mathbb{R}^n - 0, p)$. Since the fiber is homotopy equivalent to S^{n-1} , there arises a characteristic class $C(\tau_X) \in H^n(X; \mathbb{Z})$. (It is easy to understand that the fibration τ_X is simple if and only if the manifold X is orientable.)

Proposition 2. $\langle C(\tau_X), [X] \rangle = \chi(X)$.

Proof. A section of the fibration τ_X is the same as a nowhere vanishing vector field on X . It is easy to understand that a generic vector field on X has only isolated zeroes. Take a local coordinate system with the origin at the isolated zero x_0 of a vector field ξ , take a small sphere $S \approx S^{n-1}$ centered at x_0 , and consider the map $S \rightarrow S^{n-1} \rightarrow S^{n-1}$ which takes $x \in S$ into $\xi(x)/\|\xi(x)\|$. Denote by $d_\xi(x_0)$ the degree of this map. We can assume (although it is actually not necessary) that all the zeroes of ξ are *nondegenerate*, that is, $d(x_0) = \pm 1$. Now consider a smooth triangulation of X such that all zeroes of ξ lie inside n -dimensional simplices, at most one in every simplex. Then ξ is a section of the fibration τ_X over the $(n-1)$ st skeleton of X , and the obstruction c_ξ to extending this section to an n -dimensional simplex s is zero if s does not contain zeroes of ξ and is $d(x_0)$ if s contains a zero x_0 of ξ . Since the fundamental cycle of $[X]$ is the sum of all (oriented) n -dimensional simplices of the triangulation, $\langle c_\xi, [X] \rangle = \sum_{x_0 \in \{\text{zeroes of } \xi\}} d(x_0)$. The left-hand side of this equality is $\langle C(\tau_X), [X] \rangle$, the right-hand side, as explained in Sect. 17.6 (see Theorem 3 and the discussion after it), is $\chi(X)$. This completes the proof of Proposition 2.

Corollary. A connected closed orientable manifold possesses a nowhere vanishing vector field if and only if $\chi(X) = 0$.

The only if part of this statement has been proved before: See Theorem 2 in Sect. 17.6. The if part was promised there. The orientability condition is not needed; it also was explained in Sect. 17.6.

In conclusion, a couple of additional exercises.

EXERCISE 8. Make up a theory of obstructions to extending sections in the context of Serre fibrations (see Sect. 9.4).

EXERCISE 9. Let X be a CW complex with $\pi_0(X) = \pi_1(X) = \cdots = \pi_{n-1}(X) = 0$, $\pi_n(X) \neq 0$. Prove that the characteristic class of the Serre fibration $EX \rightarrow X$ with the fiber ΩX which belongs to $H^n(X; \pi_{n+1}(\Omega X)) = H^n(X; \pi_n(X))$ is just the fundamental class of X .